ESKWELABS INNOVATION FELLOWSHIP

Data Innovation Track

EIF Final Culmination

The Story of My Student Retention Synthetic Dataset Project

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What this deck is

"What did I do, and what does it say about me?"

Projects → Portfolio → Story. My EIF work is a single Synthetic Dataset project, delivered in three parts:

- **D1** a **whitepaper** that specifies a two-layer model of student dropout,
- **D2** a **generator** and a 50,000-student dataset built from it,
- **D3** an **analysis** that grades common methods against a known true effect.

This deck is a reflection on work that already exists and a map to its files. It is not a new project, and it makes no claim of being perfect.

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Everything numerical here is synthetic

Synthetic data notice. Each figure in this deck describes the behaviour of a generative model calibrated to cited *aggregate* statistics. None of it measures any real Philippine student or institution, and no number should be reported as an empirical finding. The only real numbers are model inputs tagged `CALIBRATED` with a citation.

Every parameter carries one of three provenance tags:

LOCKED
fixed by design

CALIBRATED
set to reproduce a cited statistic

ASSUMED
a stated modelling choice, no citation

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Who I am

- I am **Matthew C. Batitay**, a Fellow in the **Eskwelabs Innovation Fellowship (EIF)**, Data Innovation Track.
- My work lives on GitHub as `machew-b`, in one public repository: Student Retention Synthetic Data Generator.
- In the fellowship I took on a **Synthetic Dataset** project: model a real-world system, encode its variables and relationships mathematically, and generate data with realistic correlations, drop-off and edge cases.
- I chose student retention in Philippine higher education: published attrition figures exist only in aggregate, and records show what a student does but not why a student leaves.

How I work

Four habits run through all three deliverables, and each can be checked in the repository rather than taken on my word.

- **Write every assumption down.** All 60 generator parameters carry a provenance tag and are logged in a file. Nothing enters the model untagged.
- **Build in a known answer.** I fixed the true effect of the support programme by design, so any analysis can be scored against the truth rather than only judged plausible.
- **Separate what matched by construction from what was independently checked.** Fitting to a target shows only that a loop converged. I keep that apart from checks with independent content.
- **Say what the work is not.** The dataset ships with a usage notice, every output carries a synthetic-data banner, and the deck has a slide titled "what these results are not".

Skills snapshot

Area	What I did with it	Where to see it
Modelling and simulation	Latent-variable panel model, discrete-time hazard, nested-bisection calibration	Whitepaper; generator
Causal inference	Mantel–Haenszel, cluster-robust Poisson, inverse-probability weighting, DAG	Analytics, Sec. 8
Predictive modelling	Grouped cross-validation, logistic hazard, boosted trees, leakage guard	Analytics, Sec. 6
Python data stack	numpy, pandas, SciPy, statsmodels, scikit-learn, matplotlib	Both notebooks
Reproducibility	Seeded random streams, SHA-256 of every table, pinned requirements	Generator; analytics Sec. 3
Research and sourcing	Traced attrition figures to their original reports and cohorts	Whitepaper, Sec. 10
Technical writing	29-page whitepaper, READMEs, 19-page LaTeX slide deck	All three folders
Responsible data practice	Synthetic notice, no region names, regional tier kept out of the flagging model	Whitepaper, Sec. 13

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The track, the brief, and my three deliverables

The Data Innovation Track's Synthetic Dataset project has three deliverables. I met each with one artefact.

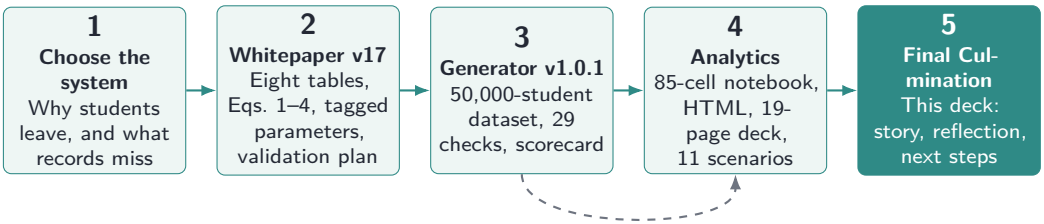
	Deliverable asked for	What I delivered
D1	Data model document with 5+ tables and latent and observed relationships	Whitepaper (v17), 8 tables, hidden and recorded layers, 29 pages
D2	Notebook that generates a customisable number of rows across the tables	Generator notebook v1.0.1 and the 50,000-student dataset it produced
D3	Notebook and/or presentation showcasing analytics on the dataset	Analytics notebook (85 cells), executed HTML export and a 19-page slide deck

What "done" meant, and how I met it

Definition of done	How I met it
Notebook runs end to end from a clean environment	Library versions are pinned in a `requirements.txt` in each notebook folder. The generator runs in about 30 to 50 seconds at 50,000 students, and the analysis in about 10 to 15 minutes.
Every assumption and distribution choice is written down	60 generator parameters (51 ASSUMED, 6 CALIBRATED, 3 LOCKED) in `parameter_log.csv`; the analysis adds its own tagged registry; limitations are logged as open flags OF-1 to OF-12 and DF-1 to DF-9.
Data survives a sanity check, with at least two deliberate edge cases	29 integrity checks plus a correlation test; two engineered edge cases (2,937 and 774 student-terms).
A short README states what a user could analyse	The generator README lists five uses, from dropout prediction to estimator-choice exercises.

The path through the work

The work did not run in a straight line. The specification came first, the generator exposed where the specification and its implementation disagreed, and the analysis then stress-tested those disagreements instead of hiding them.



The good and the bad

The good

The analysis notebook carries a verbatim copy of the generator engine and regenerates all eight tables bit for bit. Seven of seven comparable scorecard rows re-derive identically from the CSV files.

The true effect is known (30%), so methods can be graded: the naive comparison says support *raises* dropout by 57.7%, while Mantel–Haenszel gives a 32.2% reduction (95% CI 24.3% to 39.3%).

The bad

The whitepaper's design targeting (70% sensitivity, 25% false flags) is not what the generator realises (about 16% and 22%). I did not re-fit the assignment parameters to force a match (OF-1).

The held-out check passes only narrowly: 1.673 against a band of 1.2 to 1.7, with about 60% of 20 replicate populations inside the band. Inverse-probability weighting swings from -47% to $+72\%$.

The old and the new

The old

The whitepaper's Appendix A reports a *pilot* implementation (Mantel–Haenszel 25.5%, held-out ratio 1.72, not passed).

Generator v1.0.0 kept its frozen calibration constants rounded and left some structural counts untagged.

The new

Version 1.0.1 stores the calibration at full precision and tags every number. The released dataset's numbers supersede the pilot's.

The analysis adds eleven recalibrated sensitivity scenarios, which answer the question OF-1 left open, and a plain statement of what the results are not.

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The project at a glance: the problem

My EIF work is one project with three deliverables. Each is shown the same way: **Problem / Context**, **My role and contributions**, **Output**, and **Skills demonstrated**.

Problem / Context

- Philippine tertiary records show what a student does (grades, attendance, financial-account status) but not why a student leaves, and no public dataset links the two at the individual level.
- National and regional attrition figures exist, but they are aggregates and do not all trace to the same cohort.
- Early-warning systems tend to base support on the same signals that predict dropout, so the effect of a programme cannot be read off a naive comparison. Reviews call this the actionability gap (Lopez-Muñoz et al., 2026).

The project at a glance: the idea

I treated the gap between recorded and true state as the thing to model. A **hidden layer** (baseline vulnerability v_i and a per-term acute shock) drives both the recorded data and the dropout, and is never exported. Support is assigned from *prior-term* signals, and the true effect is fixed at a risk ratio of exactly 0.70 (`LOCKED`, a design constant, not an empirical estimate). With σ the logistic function:

$$\text{risk}_{it} = \sigma\big(\beta_0 + \beta\, k_{it}\big), \quad k_{it} = \mathbb{1}(\text{grade}_{i,t-1} < \tau_g) + \mathbb{1}(\text{attend}_{i,t-1} < \tau_a) + \text{hold}_{i,t-1},$$
$$h_{it} = \sigma\big(\gamma_0 + \gamma_1 v_i + \gamma_2\, \text{shock}_{it}\big) \times \big(1 - 0.30 \cdot \text{support}_{it}\big).$$

Here k_{it} counts the prior-term signals that fire. Support is drawn as Bernoulli(risk_{it}) from term 2 onward, so it depends on observed signals that also predict dropout; dropout is drawn as Bernoulli(h_{it}) with the true effect built into the second line.

Everything downstream is a test of whether an analyst can **recover that effect**.

Deliverable 1, Whitepaper (1/2): context and my role

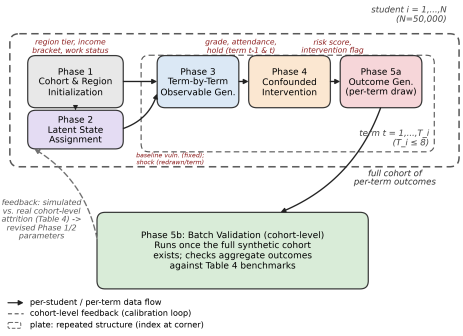
A Two-Layer Synthetic Data Generator for Studying Student Retention Early-Warning Systems in Philippine Higher Education. 29 pages, version 17.

Problem / Context. A data model with at least five tables and an explicit latent-to-observed relationship. Published attrition figures disagree, no individual-level mechanism data is public, and a model that cannot be checked is just a story.

My role and contributions

- Designed the eight-table schema (`Regions`, `Students`, six student-by-term panel tables) and a five-phase pipeline for 50,000 students over up to eight terms (`LOCKED`).
- Wrote the generative forms, including the confounded assignment rule and the fixed 30% effect: a built-in answer key.
- Traced the calibration targets to their sources (CPBRD, 2025; EDCOM II, 2025; Commission on Higher Education, 2025; Aldaba et al., 2024). Calibrated to the Class-of-2022 tier rates (67.1% high, 45.9% typical, `CALIBRATED`) and held a 29.4% national figure out for an independent check.

Deliverable 1, Whitepaper (2/2): the rest, output and skills



The end-to-end pipeline (whitepaper Figure 1). Batch validation (5b) feeds back into calibration (dashed arrow).

More of my role

- Wrote a validation plan that keeps calibration-until-convergence apart from independent checks (Law, 2015): held-out, confounding, recoverability.
- Specified two deliberate edge cases, a responsible-use section, and a field-level schema with integrity rules (Appendix B).

Output

Whitepaper PDF and LaTeX source in `whitepaper/`.

Skills demonstrated

Schema and hazard modelling; tracing sources; validation design (Law, 2015; Gelman et al., 2020); LaTeX writing; research ethics.

Deliverable 2, Generator (1/3): context and my role

Generator v1.0.1. One notebook (36 cells), eight tables, 297,739 rows in each panel table.

Problem / Context. A specification only counts if it can be run. D2 asks for a notebook that generates a customisable number of rows across the tables. The calibration targets are cohort-level, so the generator has to find its own hazard parameters.

My role and contributions

- Implemented the generator from the whitepaper. One configuration cell sets the number of students, the master seed, recalibration and the output folder.
- Built a parameter registry: every number is read through one function and carries a tag (60 parameters: 51 `ASSUMED`, 6 `CALIBRATED`, 3 `LOCKED`).
- Drew every random quantity from its own named stream under one master seed, with common random numbers so the calibration loop is stable. A nested bisection fits the tier shift and hazard intercept to the two tier targets.

Deliverable 2, Generator (2/3): the rest, output and skills

More of my role

- Wrote 29 integrity checks and a correlation test, a scorecard for every check in the whitepaper's validation table, an edge-case audit, and open flags OF-1 to OF-12.
- Shipped v1.0.1 after finding that v1.0.0 kept its frozen calibration rounded and left some counts untagged, so the default dataset reproduces exactly.
- Withheld the hidden state: baseline vulnerability, shocks and hazard are never exported, and no individual-level table carries a region name.

Output. Eight CSV tables, `metadata.json` with the SHA-256 of every table, the parameter log, the validation scorecard, a dataset notice and diagnostics, in `synthetic-data-generator/`.

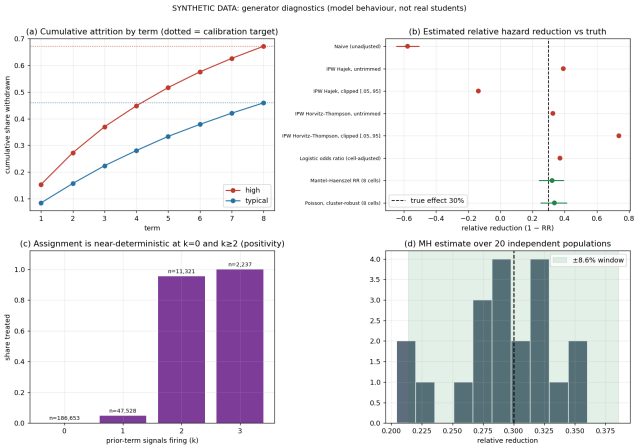
Skills demonstrated. Simulation engineering in Python; calibration by nested root-finding; reproducibility (named streams, common random numbers, content hashes); automated data validation; documenting assumptions and limits; privacy-aware dataset design.

Deliverable 2, Generator (3/3): the scorecard

The released dataset (synthetic; model behaviour, not reality). The first two rows are calibration, not independent validation.

Check	Result	Verdict
Tier attrition, high / typical	67.1% / 45.9% (calibrated targets)	Pass
Pooled attrition (derived)	49.1%	Pass
Held-out tier ratio, band 1.2 to 1.7	1.673; over 20 populations 1.692 ± 0.025, about 60% in band	Pass (fragile)
Confounding present	Naive estimate -57.7% (CI -64.6% to -50.6%) excludes the true 30%	Pass
Effect recoverable	Mantel–Haenszel 32.2% (CI 24.3% to 39.3%); Poisson 33.6% (CI 25.1% to 41.2%)	Pass
Parameter traceability	60 parameters, none untagged	Pass
Edge cases present	2,937 and 774 student-terms	Pass

Deliverable 2, Generator: the diagnostics figure



Reading the four panels (synthetic)

(a) Cumulative attrition by term against the calibration targets.

(b) Each estimator against the true 30%.

(c) Assignment is near-deterministic when no signal or at least two signals fire.

(d) The Mantel–Haenszel estimate over 20 independent populations.

Deliverable 3, Analytics (1/3): context and my role

Analytics notebook and slide deck. 85-cell notebook, executed HTML export, 19-page deck (17 slides and 2 backup).

Problem / Context. D3 asks for analytics on the dataset. I wanted more than a tour of charts: because I know the truth behind the data, I could ask whether the workflow a Student Success Analyst would normally follow reaches the right conclusion, and where it does not.

My role and contributions

- Wrote the analysis so it sees only what an analyst would see. The hidden state is used only in labelled places, always to *grade* a result.
- Copied the generator engine unchanged into the notebook and proved the copy faithful: it regenerates all eight tables with matching hashes.
- Built the prediction work with a leakage guard, student-grouped five-fold cross-validation, fixed hyperparameters, and two ceilings.

Deliverable 3, Analytics (2/3): the rest, output and skills

More of my role

- Kept the regional tier out of the flagging model on ethical grounds; an ablation shows it adds almost nothing (AUC 0.656 with it, 0.655 without).
- Ran the causal analysis with several estimators over 20 independent populations, an adjustment-set ladder and weight diagnostics (Peters et al., 2017; Gelman et al., 2020).
- Audited targeting and ran 11 recalibrated sensitivity scenarios. Wrote eight recommendations, a limitations log (DF-1 to DF-9) and the deck.

Output. Analysis notebook, executed HTML export, and the slide deck with its LaTeX source, in `results-and-discussion/`.

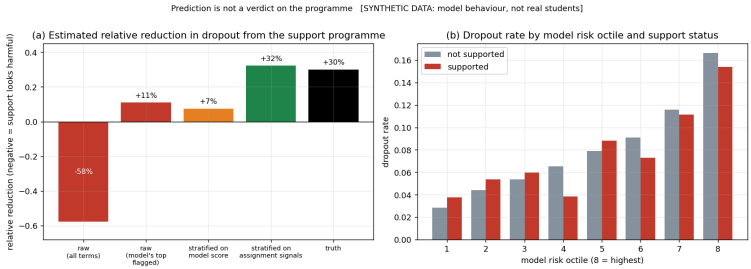
Skills demonstrated. Applied causal inference; predictive modelling with leakage control (James et al., 2013); sensitivity analysis; reproducible notebooks; storytelling for two audiences; stating limits plainly.

Deliverable 3, Analytics (3/3): what it found

Synthetic; properties of the generator, not of real students.

Question	Headline
Is the generator valid?	Seven of seven comparable scorecard rows re-derive identically from the CSV files.
Can dropout be predicted?	Out-of-fold AUC 0.655 for boosted trees against 0.567 for a signal-count rule. A predictor that knew every student's hidden vulnerability would reach 0.683. A further 19.4% of dropouts happen in term 1, before any prior record exists.
Does a good predictor say whether support works?	No. Raw comparison: support appears to raise dropout by 58%. Stratifying on the model's risk score: +7.4%. Stratifying on the assignment signals: +32.2% against a true 30%.
Does the programme work?	Mantel–Haenszel and Poisson recover about 30% (SD about 4 points over 20 populations). Inverse-probability weighting ranges from −47% to +72%.
Who is missed?	63.7% of dropout events fall in terms with a clean record and no support. A model-ranked list would avert about 18% more dropouts than the actual rule at equal capacity.
Is it robust?	Across 11 recalibrated scenarios the Mantel–Haenszel estimate stays near 30%.

Deliverable 3, Analytics: the actionability gap



(a) Five ways of estimating the programme's effect, against the true 30%. (b) Dropout rate by model risk octile and support status. Synthetic.

The raw comparison says support hurts (−58%). Stratifying on the model's own score gives +7%. Stratifying on the **assignment signals** recovers +32%, against a true +30%.

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What did I learn?

- **Matching a target is not validating a model.** Fitting two parameters to the tier rates shows only that the calibration loop converged. The checks with independent content are the correlation structure, the confounding check and the recoverability check, and I report them separately.
- **A good predictor is not a verdict on a programme.** The boosted model has a modest but real AUC of 0.655, yet stratifying on its own risk score gives $+7.4\%$ where the truth is 30%. Knowing the assignment rule is what makes the comparison honest.
- **Estimator choice can matter more than sample size.** With 50,000 students and the true propensity supplied, inverse-probability weighting still ranged from -47% to $+72\%$, because assignment is near-deterministic in some groups. Methods that borrow strength across groups had a standard deviation of about 4 points.
- **An assumptions list is a design tool.** The 51 parameters tagged `ASSUMED` became a ready-made agenda for sensitivity analysis.

What challenged me?

- **Results that did not match my own design.** The whitepaper states 70% targeting precision and a 25% false-flag rate; the generator realises about 16% and 22%. The pair is not jointly reachable under the chosen rule. The temptation was to tune until the numbers matched. I held to the rule that a pre-specified quantity is not changed to make a check pass: I disclosed it, labelled it, and moved the question to the sensitivity analysis.
- **A check that passes by a hair.** The held-out tier ratio sits at the upper edge of its band, so I labelled it "pass (fragile)".
- **Sources that disagree.** National attrition figures of roughly three and four in ten do not trace to the same cohort. One reading (that a 29.4% figure belongs to the Class of 2023) is still an inference from a table layout, flagged as open (OF-7).
- **Keeping synthetic data from being mistaken for findings.** Every output carries a banner, individual-level tables carry no region names, and the recommendations end with a plain instruction not to act on these numbers.

What am I proud of?

- The project can be checked. A reader can regenerate all eight tables bit for bit, re-derive the scorecard from the CSV files, and see where each number came from.
- The deliverables form one argument: the whitepaper states a model, the generator realises it, and the analysis tests what a standard workflow would conclude from its output.
- The limitations are in the open. Twelve generator flags and nine analysis flags are logged with a status, including the ones that remain unresolved.
- The analysis is written from an analyst's point of view and only then graded against the truth, the discipline I wanted a retention analyst to have.

What would I improve?

- **Bring the whitepaper in line with the generator.** Its targeting figures (Sec. 8 and Table 4) should describe the realised behaviour, the appendix should say the near-determinism follows from the chosen parameters, the title page should carry a version line, and the pilot script that Appendix A mentions should be added or the sentence reworded.
- **Close the sourcing question** on the 29.4% figure against CHED's underlying data, if it can be obtained (OF-7).
- **Make the slide figures reproducible** by adding the script that draws them to the repository.
- **Loosen the generator's simplifications.** A single homogeneous effect, no carry-over from support into later grades, and no unmeasured assignment confounders make the causal problem easier than a real one.

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Finishing this project, and extending the science

Finishing. The first thing I would do is publish a whitepaper v18 that fixes the inconsistencies listed above. After that I would resolve OF-7, add the figure script, and re-run the held-out check that the analysis did not repeat.

Extending. The analysis notebook already names the open questions:

- **Heterogeneous effects.** Let the support effect differ by student, which breaks the assumption that ranking by risk is optimal.
- **The cost of building overlap into a rollout.** I recommend a small randomised share of referrals so every signal group has both arms, but I did not measure what that costs in support slots.
- **The boosted model across scenarios.** The sensitivity grid recomputed the rule and the ceilings, but not the boosted model's accuracy (DF-5).
- **Why stratifying on the model's score is biased.** I observed the bias but did not investigate its mechanism (DF-7, open).

What real use would require

Nothing in this repository should drive a decision about real students.

A real programme would need:

- permissioned administrative data,
- ethical review, and
- a head-to-head pilot of model-based allocation, with a randomised share of referrals.

That is the point at which the recommendations in Deliverable 3 stop being **conclusions within a model** and start being **hypotheses**.

Where to find everything

Item	Location in the repository
Repository overview	`README.md`
D1 whitepaper	`whitepaper/student-retention-data-generator-model.pdf`
D2 generator and dataset	`synthetic-data-generator/`
D3 notebook, HTML, slide deck	`results-and-discussion/`
This deck	`culminating-activity/`

This is the story of my work.

The work is imperfect, the imperfections are documented, and all of it is visible.

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